Problem-Solution Fit

Date	29 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
This canvas explains how the OptiCrop: Smart Agricultural Production Optimization Engine addresses farmers' crop selection problems by using machine learning to recommend the best crop based on soil parameters and agricultural data..

1. Customer Segment(s) [CS] • **Small and medium-scale farmers** • **Farmers with limited access to agricultural experts** • **Rural farming communities**	**8. Channels of Behavior [CH]** **Online:** • **Mobile applications** • **Websites** • **Social media** • **Agricultural portals** **Offline:** • **Agricultural officers** • **Farmer meetings**	**5. Available Solutions – Pros & Cons [AS]** **Existing Solutions:** • **Agricultural experts, consultancy services** • **Government advisory systems**
2. Problems / Pains + Frequency [PR] • **Difficulty in selecting the right crop for specific soil conditions** • **Low crop yield due to improper crop choice**	**9. Problem Root / Cause [RC]** • Lack of scientific crop recommendation systems • Insufficient knowledge of soil parameters (pH, nutrients, moisture)	**6. Customer Limitations [CL]** • Limited budget • Lack of smartphones or advanced devices (in some cases) • Low digital literacy

- **Lack of awareness about soil nutrients and fertility**
- **Frequent occurrence every farming season**

• Limited access to modern agricultural technologies

3. Triggers to Act [TR]

• Decreasing crop productivity over time

Soil nutrient imbalance

• Financial losses due to poor harvest
• Need for better farming decisions
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4. Emotions Before / After [EM]

- **Before: Confusion, uncertainty, fear of losses**
- **After: Confidence, clarity, optimism in decision-making**

10. Your Solution [SL]

OptiCrop: Smart Agricultural Production Optimization Engine

- **Uses Machine Learning algorithms to analyze soil parameters such as:**
  - **pH level**
  - **Nitrogen, Phosphorus, Potassium (NPK)**
  - **Moisture and climate conditions**
- **Recommends the most suitable crop for maximum yield**
- **Helps farmers:**
  - **Improve productivity**
  - **Reduce financial losses**
  - **Make data-driven farming decisions**

7. Behavior + Intensity [BE]

- **Farmers rely heavily on traditional methods and past experience**
- **High dependence on local advice and trial-and-error methods**

OFFLINE
Agricultural officers, local markets, farmer meetings, village communities.